

Terra Vision

Terravision revolutionizes real estate investing for beginners by offering an AI-driven platform with lifelike simulations and prediction tools to simplify market analysis and decision-making.

C241-PS263



Feature & Description



Introducing: TerraVision

TerraVision is a mobile app that can **predict house prices** based on its building area, surface area, number of bedrooms, number of bathrooms, and city. This app is currently cover various properties in **Jakarta, Bekasi, Depok, Bandung, Bogor, and Tangerang** City.

Features

House Price Prediction

KPR Calculation

Add to Favorite

Implementation

Mobile Development

Machine Learning

Cloud Computing

Background & Reason

1

Investment Analysis:

Property investors can use this application to analyze the potential profits from their investments.

2

Helping Buyers and Sellers:

Helping people to search for house prices in an area with the desired house input they live in.*

What is the reason?

Failing to Make a Plan

Not Addressing Valuation Comps Properly

Unrealistic Expectation of Profit

Underestimating Costs

Purchasing Property without Ample Reserves

Getting Poor Financing



Machine Learning Implementation

Tools used:



NumPy is a fundamental package for **scientific computing** in Python. It provides support for arrays, matrices, and many mathematical functions to operate on these data structures.

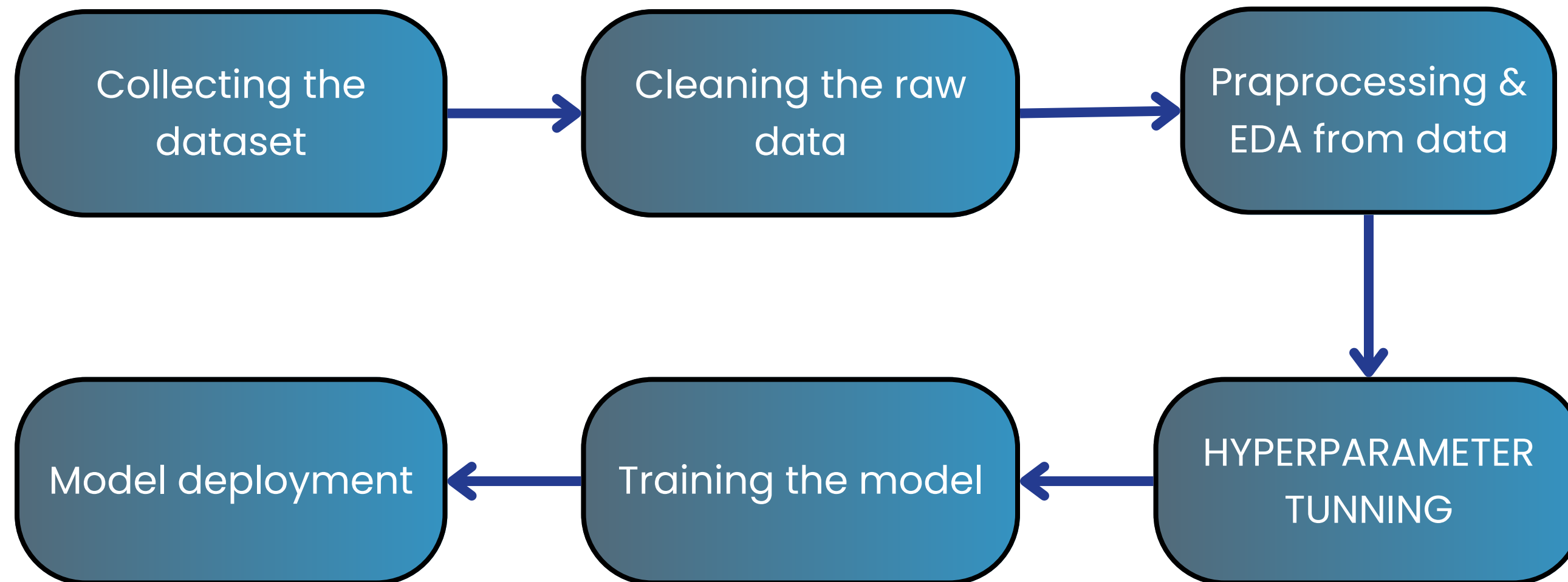


TensorFlow is an open-source **deep learning framework** developed by Google. It allows developers to create **large-scale neural networks** with many layers. and use keras-tuner for optimizing hyperparameters of machine learning,



Scikit-learn is a **machine learning library** for Python. It includes simple and efficient tools for data mining and data analysis, built on **NumPy**, **SciPy**, and **matplotlib**.

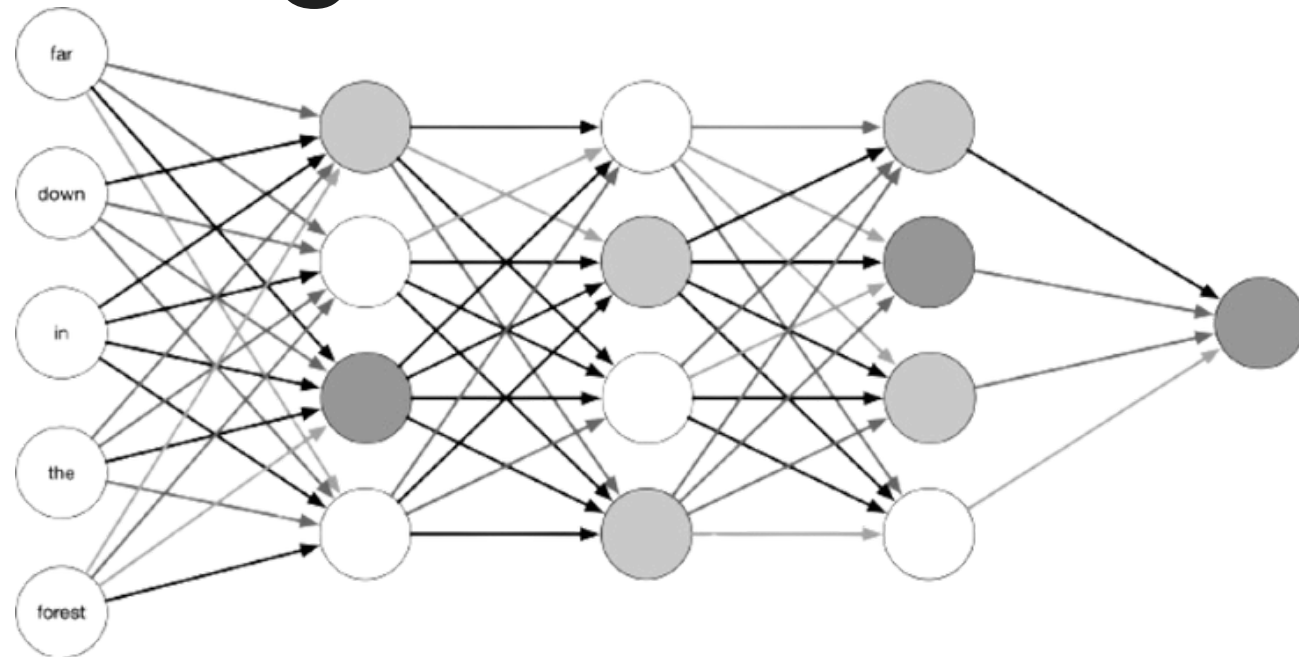
Machine Learning Implementation



Implementation

Machine Learning Documentation

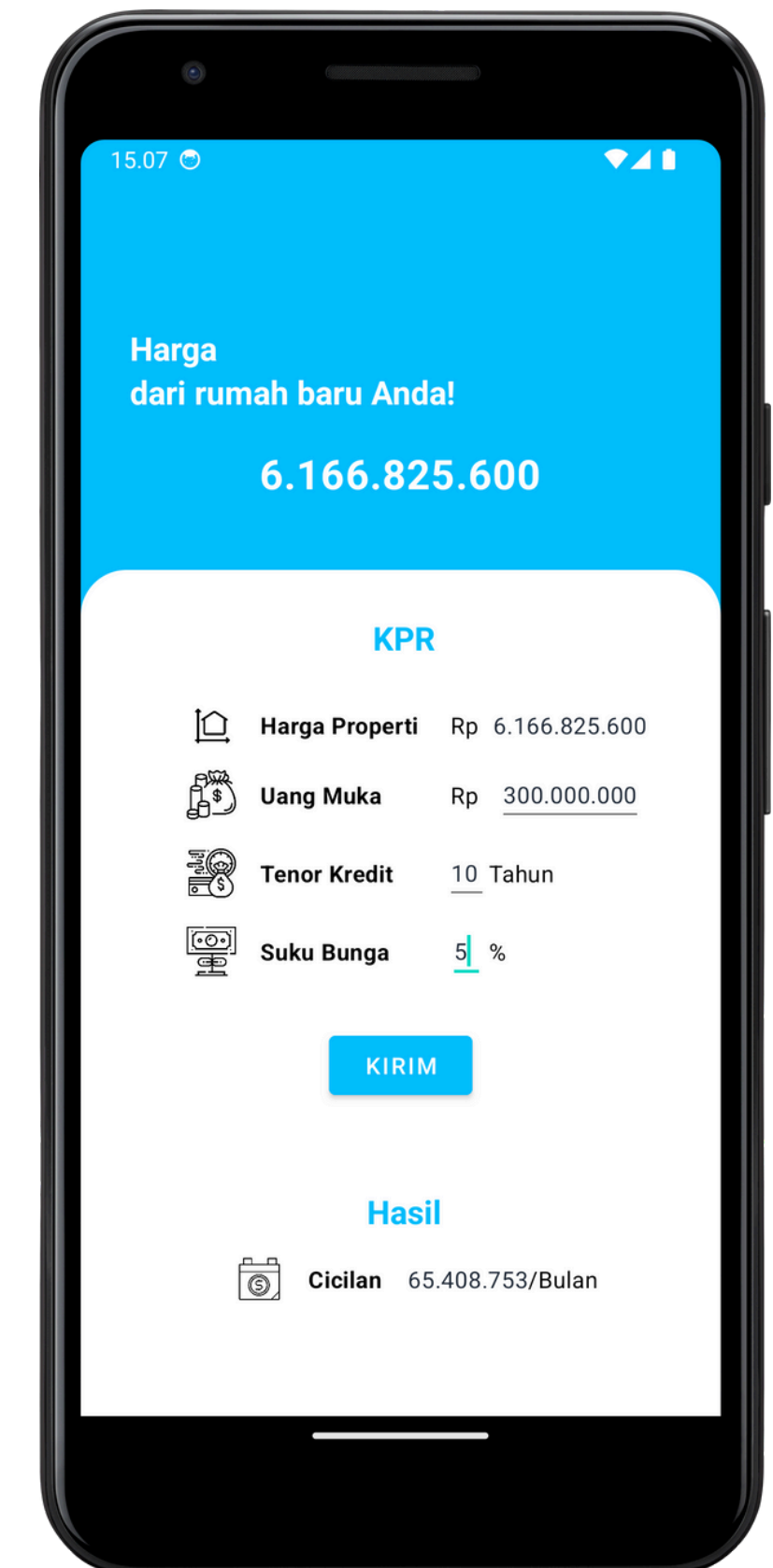
Using Dense Neural Network (DNN)



HYPERPARAMETER TUNER

```
The optimal number of units in the first densely-connected layer is 416.  
The optimal number of units in the second densely-connected layer is 192.  
The optimal number of units in the third densely-connected layer is 128.  
The optimal learning rate for the optimizer is 0.005196281231414663.  
The optimal dropout rate for the first dropout layer is 0.1.  
The optimal dropout rate for the second dropout layer is 0.30000000000000004.
```

Implementation



Machine Learning Documentation

TRAIN MODEL

```
def build_optimal_model():
    model = keras.Sequential()
    model.add(keras.layers.Input(shape=(X_train.shape[1],)))

    # Menggunakan parameter optimal yang ditemukan
    model.add(keras.layers.Dense(units=416, activation='relu'))
    model.add(keras.layers.Dropout(rate=0.1))
    model.add(keras.layers.Dense(units=192, activation='relu'))
    model.add(keras.layers.Dense(units=128, activation='relu'))
    model.add(keras.layers.Dropout(rate=0.3))
    model.add(keras.layers.Dense(1, activation='linear'))

    # Menggunakan learning rate optimal
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=0.005196281231414663),
                  loss='mean_squared_error',
                  metrics=['mean_absolute_error'])

    return model

# Membangun dan melatih model dengan parameter optimal
optimal_model = build_optimal_model()
```

API

Tools used:



Implementation

Cloud Computing

Tools used:



Google Cloud Storage



Cloud SQL

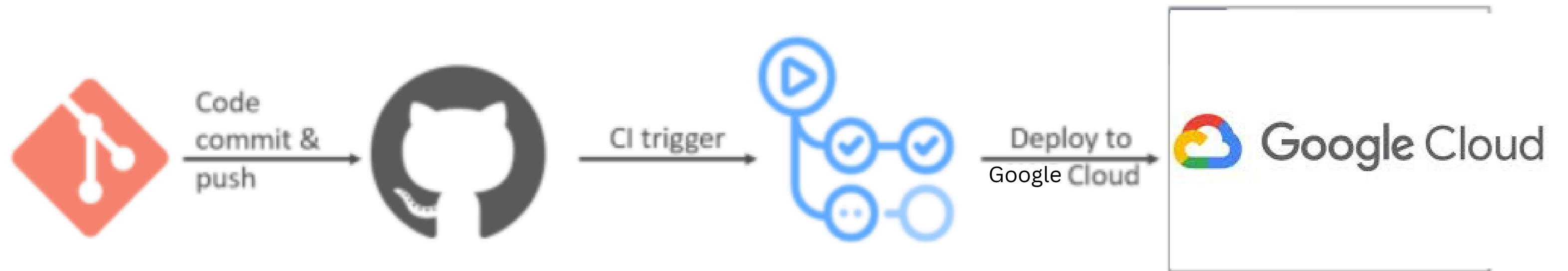


App Engine

Implementation

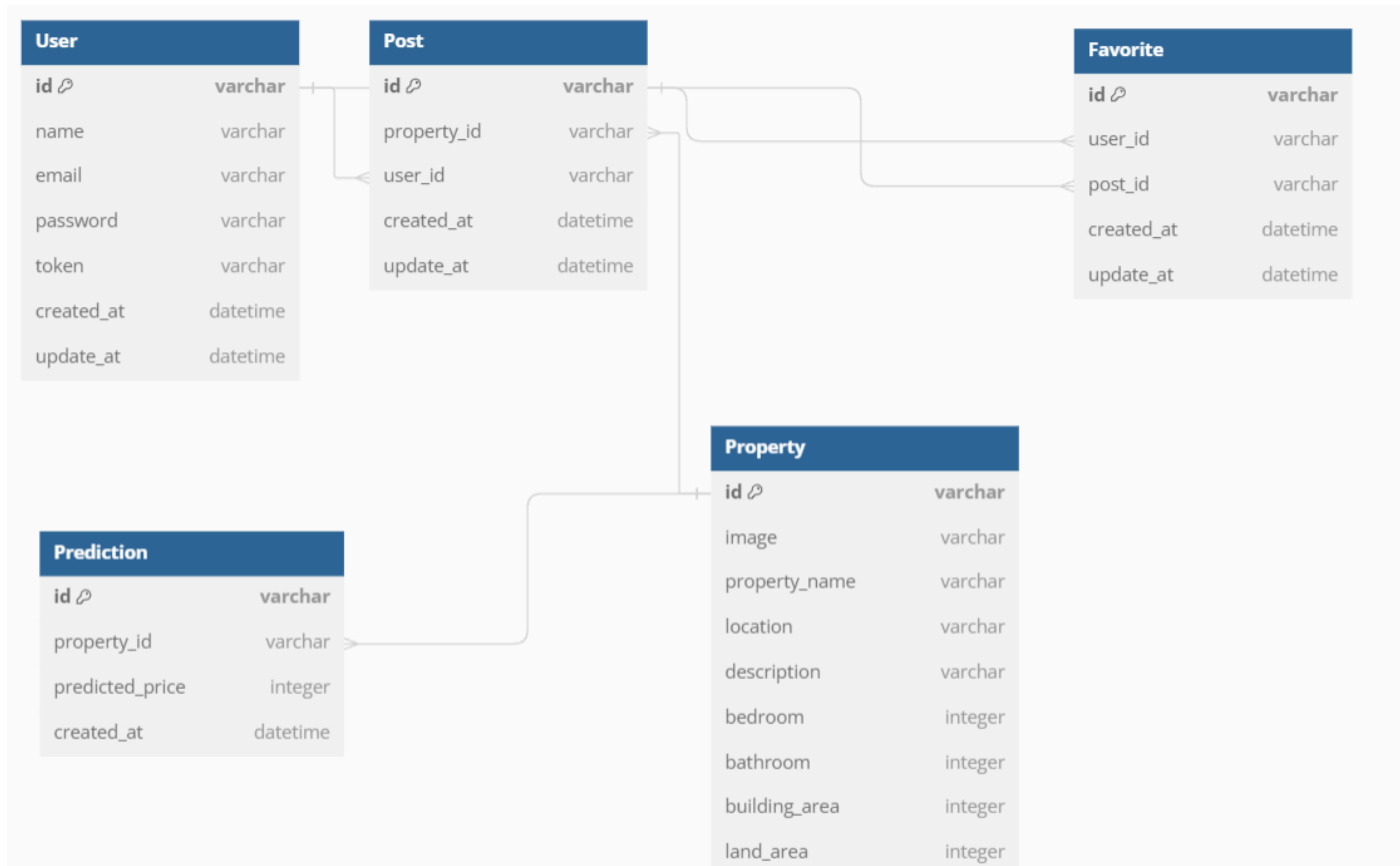
CI/CD

Tools used:



Implementation

Database Design



API Documentation

Method	Endpoint	Description
POST	/v1/auth/register	This endpoint allows users to register by providing their name, email, and password.
	/v1/auth/login	This endpoint allows users to log in and obtain authentication credentials.
	/v1/auth/logout	This endpoint is used to log out and invalidate the refresh token.
	/v1/auth/refresh-token	This endpoint is used to refresh the access token using the refresh token.
GET	/v1/post	This endpoint allows the user to create a new property listing.
	/v1/post	This endpoint retrieves a list of posts with the option to sort by year in ascending order, and paginate the results.

API Documentation

Method	Endpoint	Description
GET	/v1/post/search	This endpoint retrieves posts based on the search query and allows sorting and pagination.
	/v1/users/me	The endpoint retrieves the details of the authenticated user

Mobile Development Implementation

Tools used:

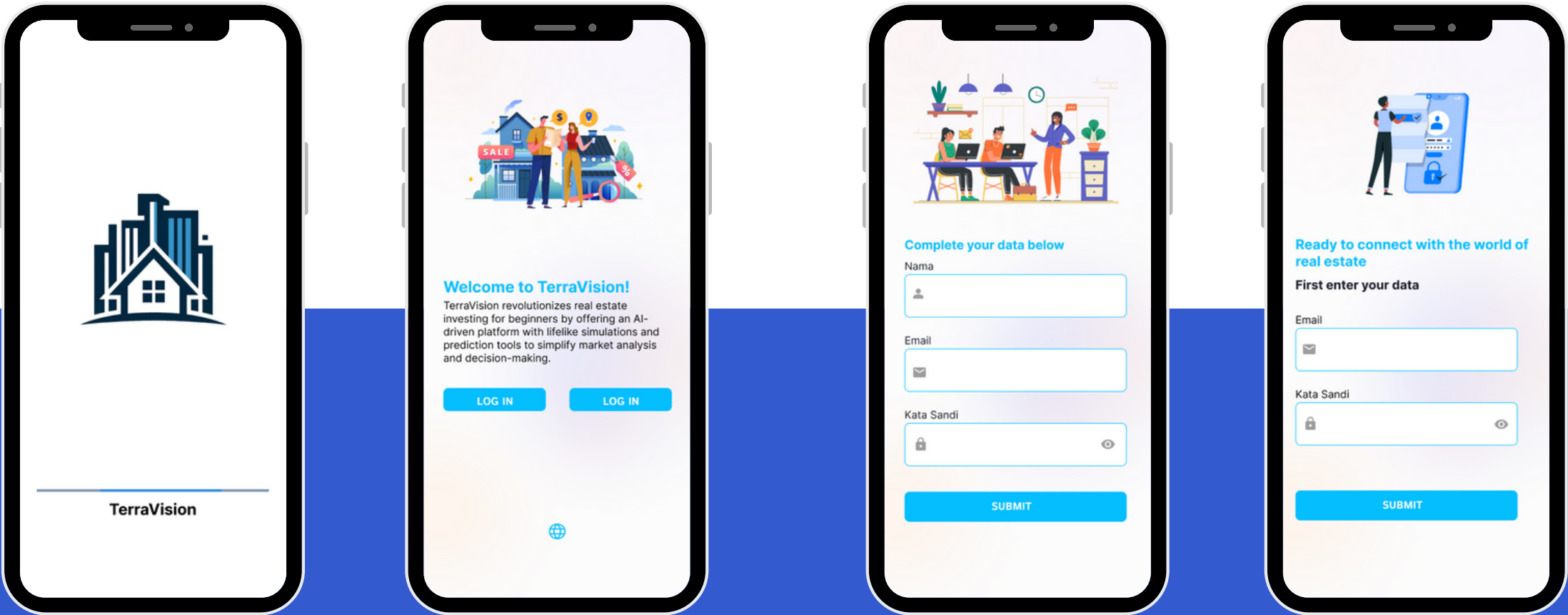


To support application development on the Android operating system, Android Studio uses a Gradle-based build system, emulator, code templates, and integration with Github.



an open-source framework from Google designed to run machine learning models with high efficiency on mobile and edge devices.

Onboarding Pages



Splash Screen - Welcome Activity - Sign Up - Sign In

Homepage

01

Profile

02

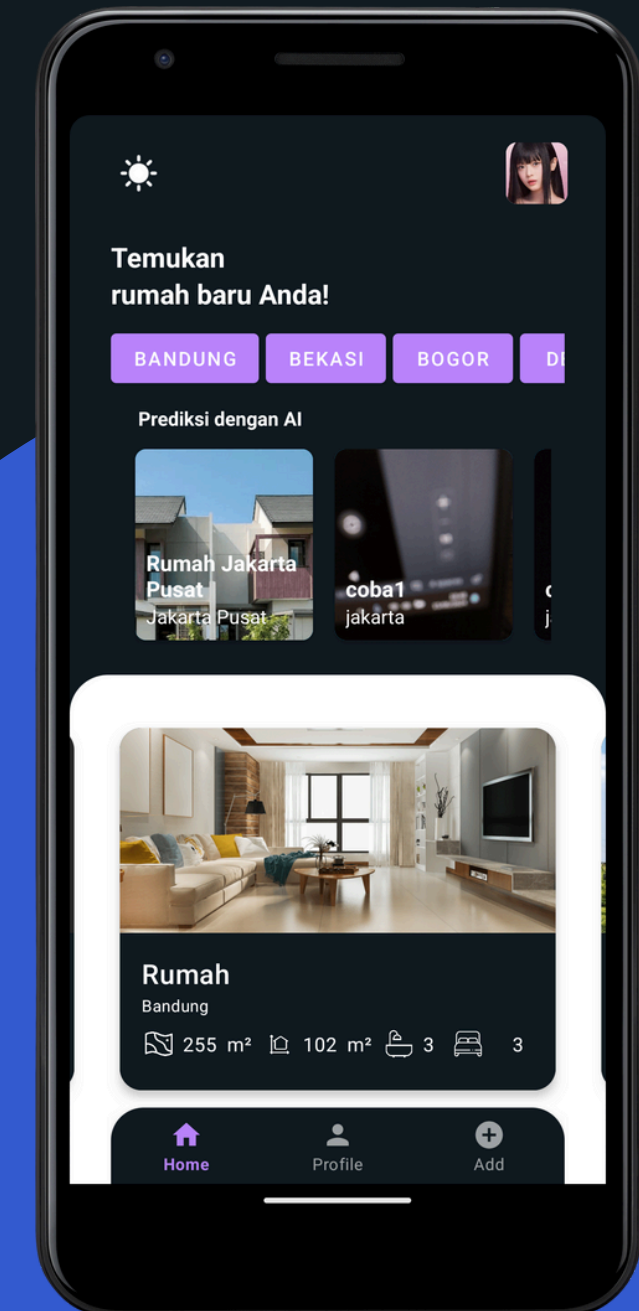
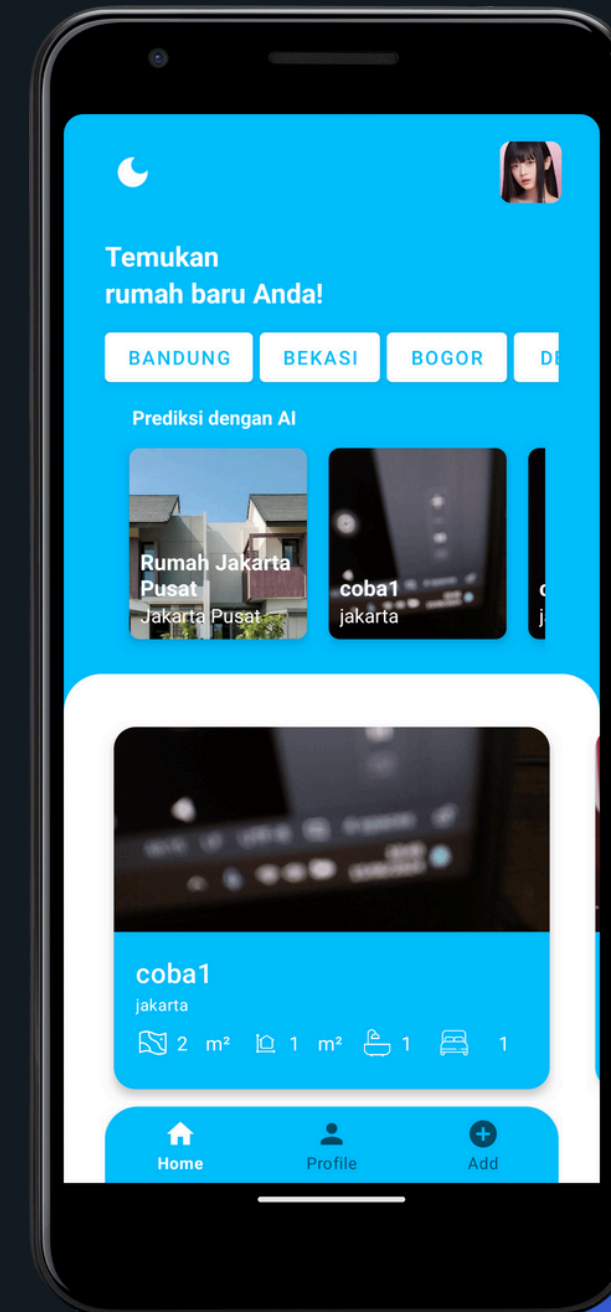
Property Location

03

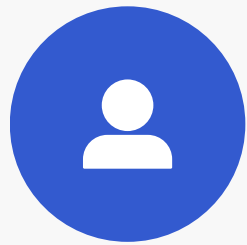
Explore

04

Add Property



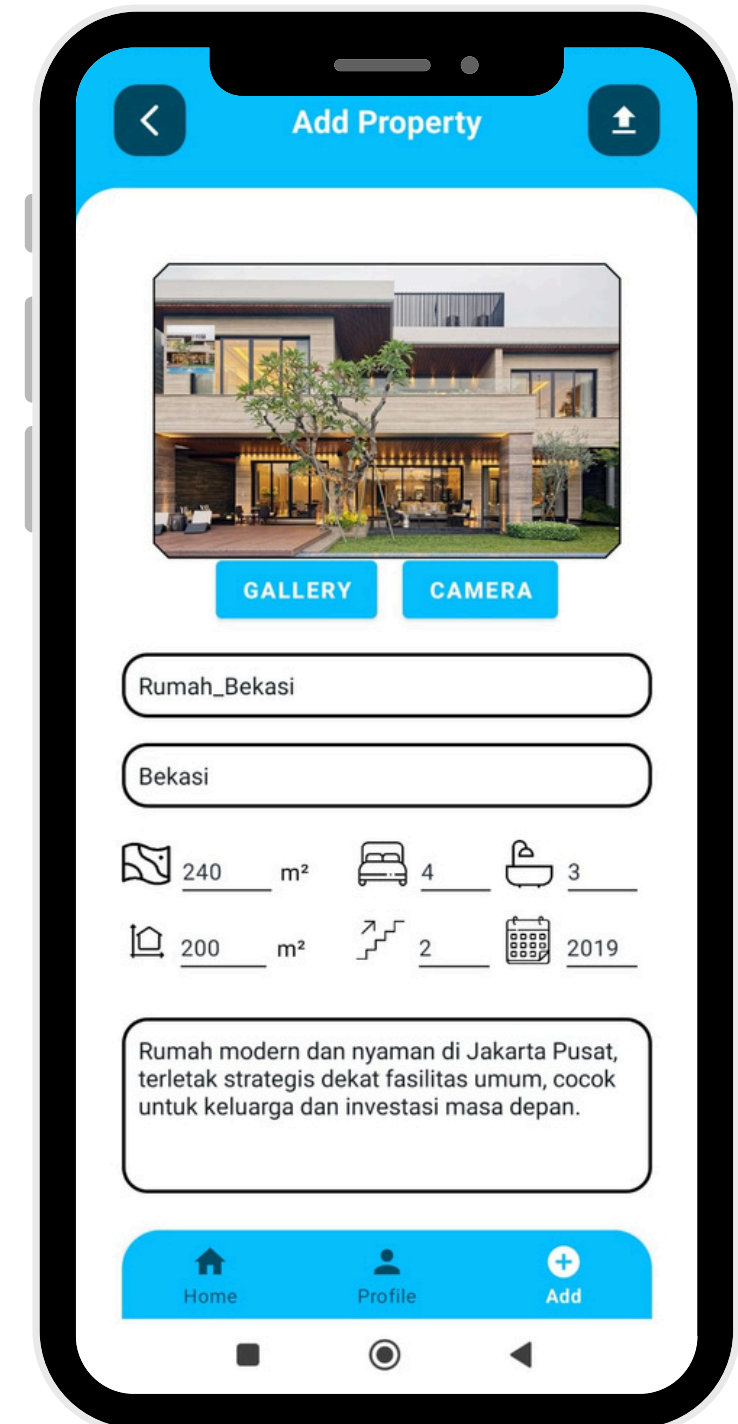
Add Property

**01**

Users can post their personal property by entering their house specifications. This allows them to predict the price of their house on their own.

**02**

Since users can post their houses, it can serve as a platform for free advertising.



Rumah Jakarta Pusat

Jakarta Pusat

Rumah modern dan nyaman di Jakarta Pusat, terletak strategis dekat fasilitas umum, cocok untuk keluarga dan investasi masa depan.

	Luas Permukaan	400 m ²
	Luas Bangunan	200 m ²
	Kamar Tidur	3
	Kamar Mandi	2
	Lantai	2
	Tahun	2019

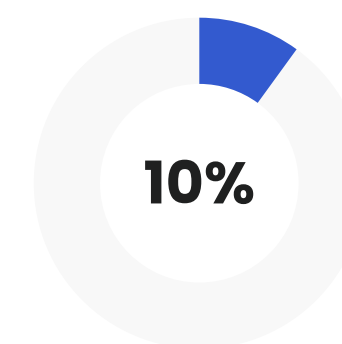


Rianco Aritonang
Aritonang@gmail.com

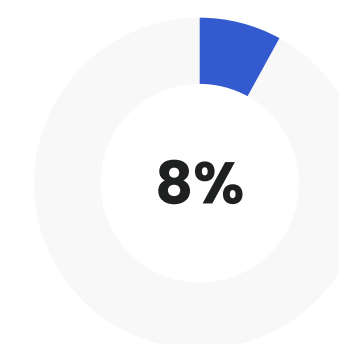
PREDIKSI HARGA



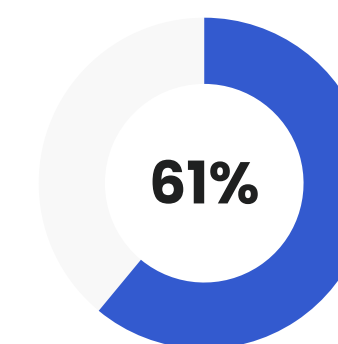
Result



Down Payment

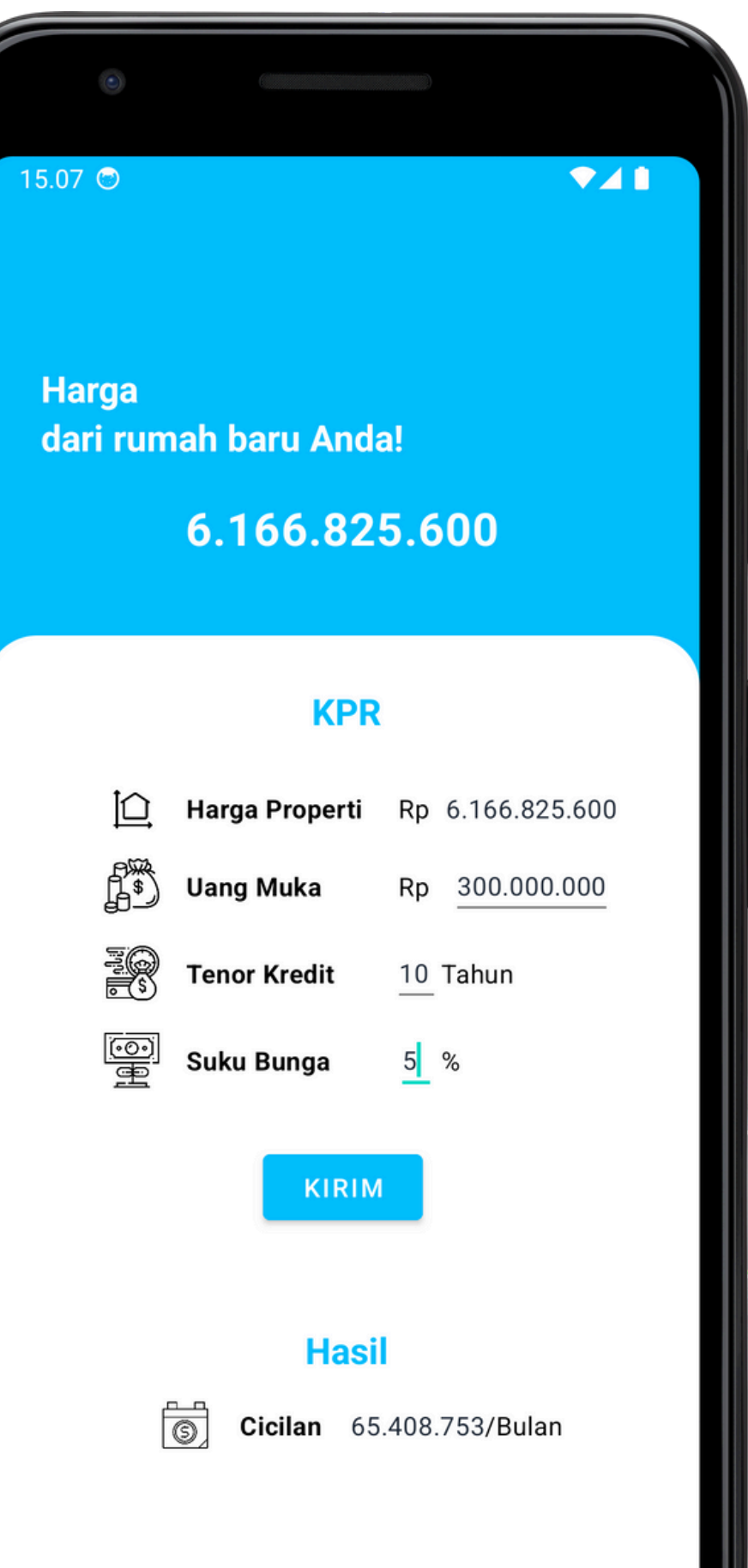


Interest Rate



Worth It

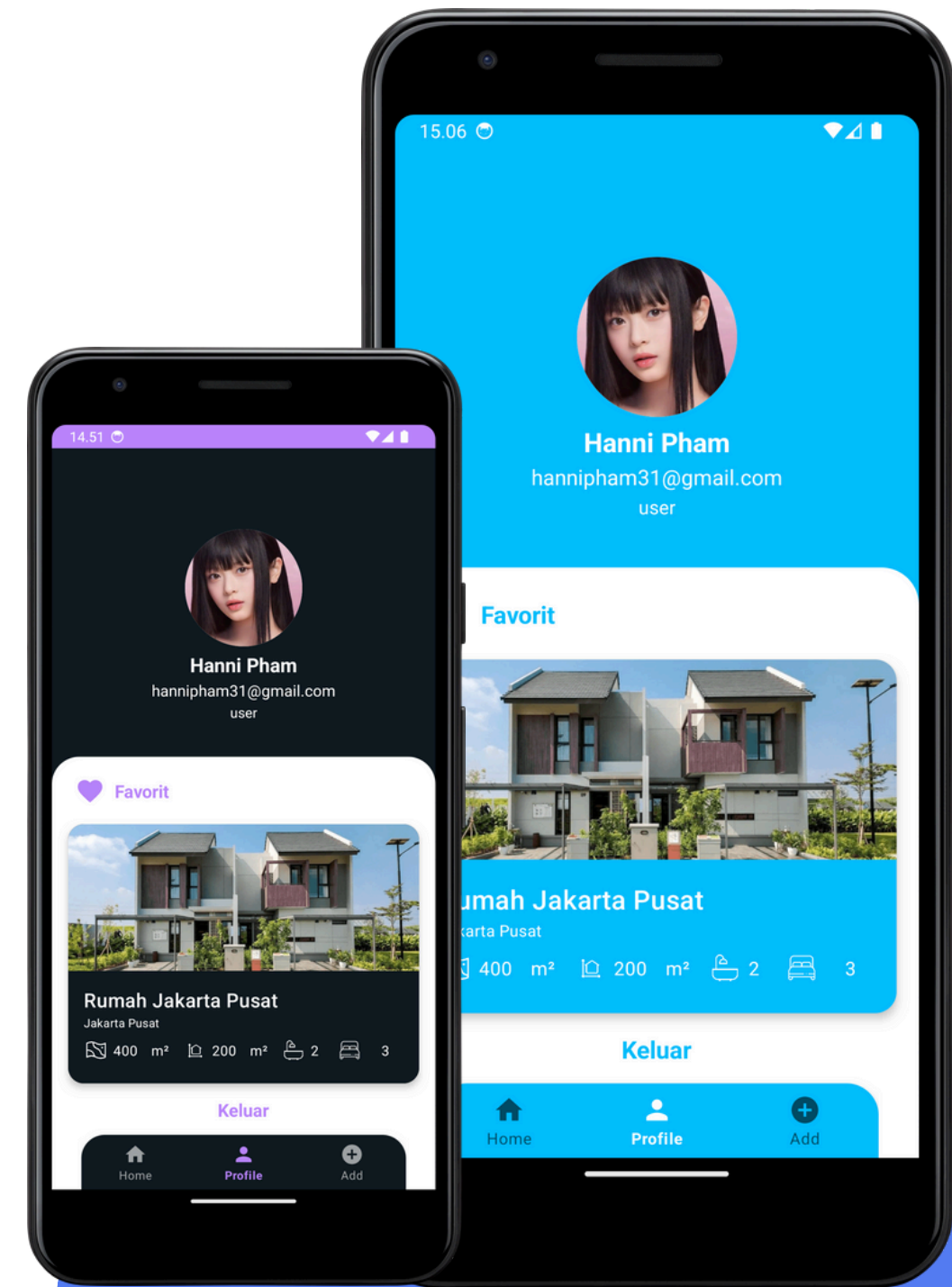
Users can calculate the monthly mortgage payments by adjusting the down payment, loan tenure, and interest rate.



Favorites Page



Users can save properties they like by pressing the favorite button and then view them again in the profile menu.



Try This App!



ristek.link/terravision

Thank You

